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(54) **DISPERSION DEVICE AND ACCUMULATION DEVICE**

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(51) **Int. Cl.**

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D01D 7/02	(2006.01)
D21B 1/08	(2006.01)

(52) **U.S. Cl.**

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(58) **Field of Classification Search**

CPC D04H 1/732; B04C 3/00; B01F 25/102; D01D 7/02

See application file for complete search history.

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(57) **ABSTRACT**

There is provided a dispersion device including: a supply section having a supply pipe for supplying a material containing fibers together with air, and a chamber that has a first swirl flow forming portion forming a first swirl flow of the air containing the material and a second swirl flow forming portion communicating with the first swirl flow forming portion and forming a second swirl flow in a direction opposite to the first swirl flow of the air containing the material and that is coupled to the supply pipe; a dispersion section that has a casing formed with a discharge port for discharging the material, stirs the material in the casing, and discharges and disperses the material from the discharge port into air; and a coupling section having a communication port through which the first swirl flow forming portion and the second swirl flow forming portion communicate with the casing.

8 Claims, 4 Drawing Sheets

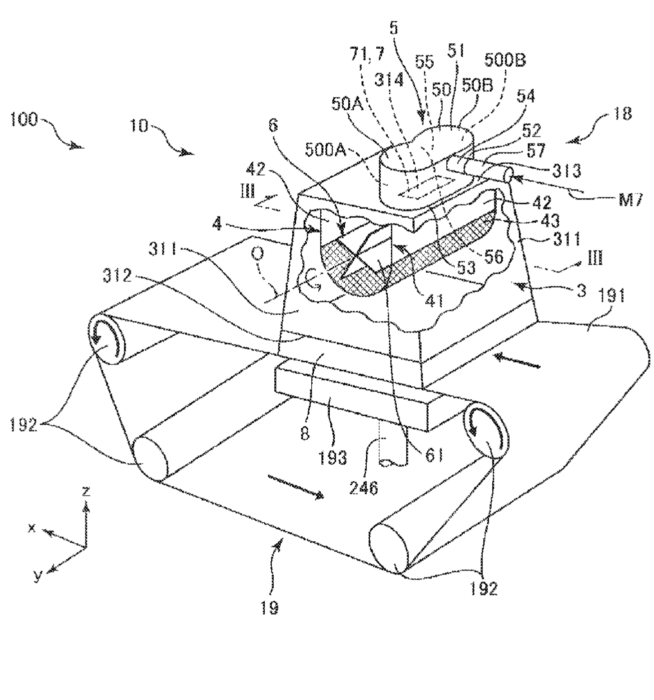

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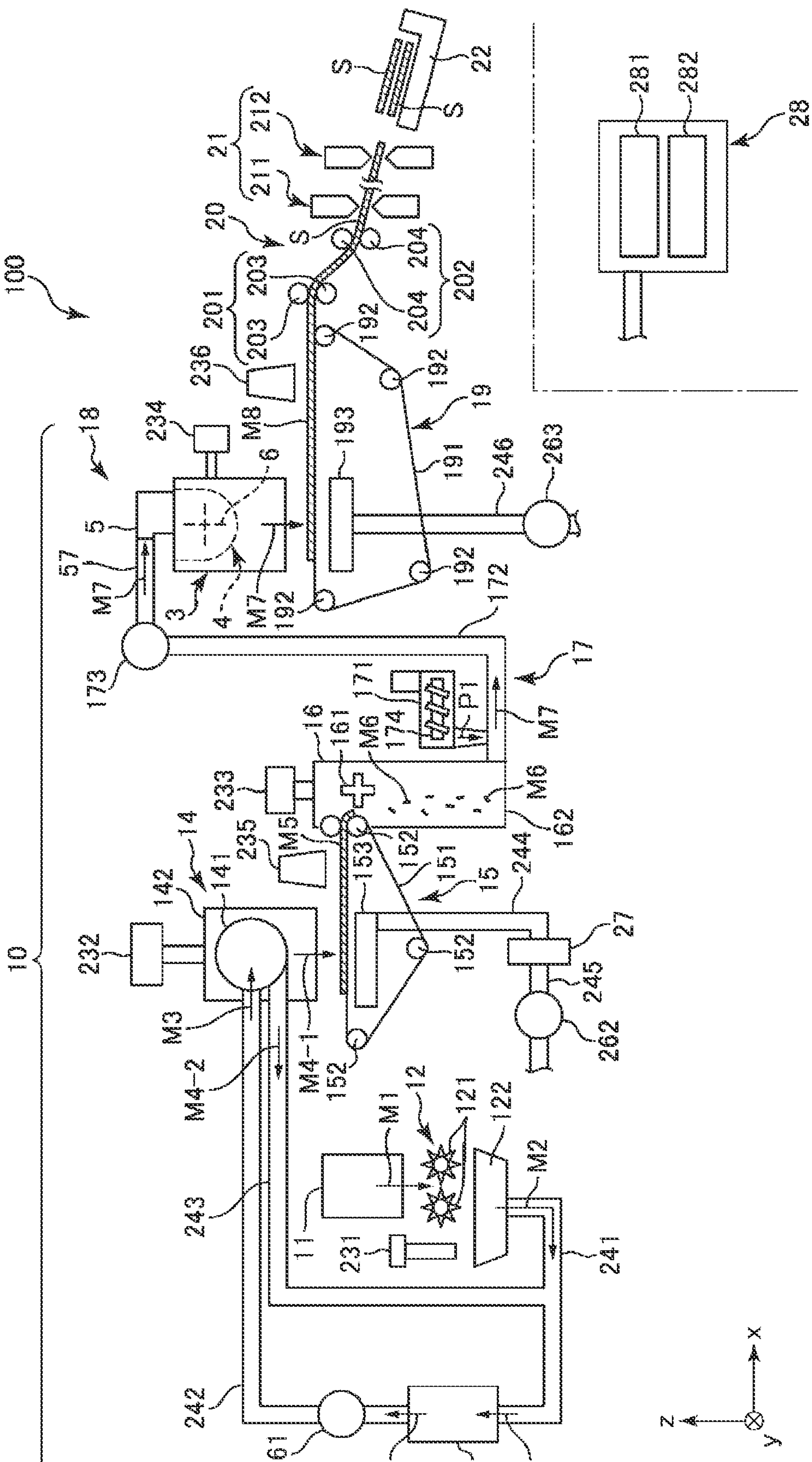
FIG. 1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P1
18
19
191
192
193
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201
202
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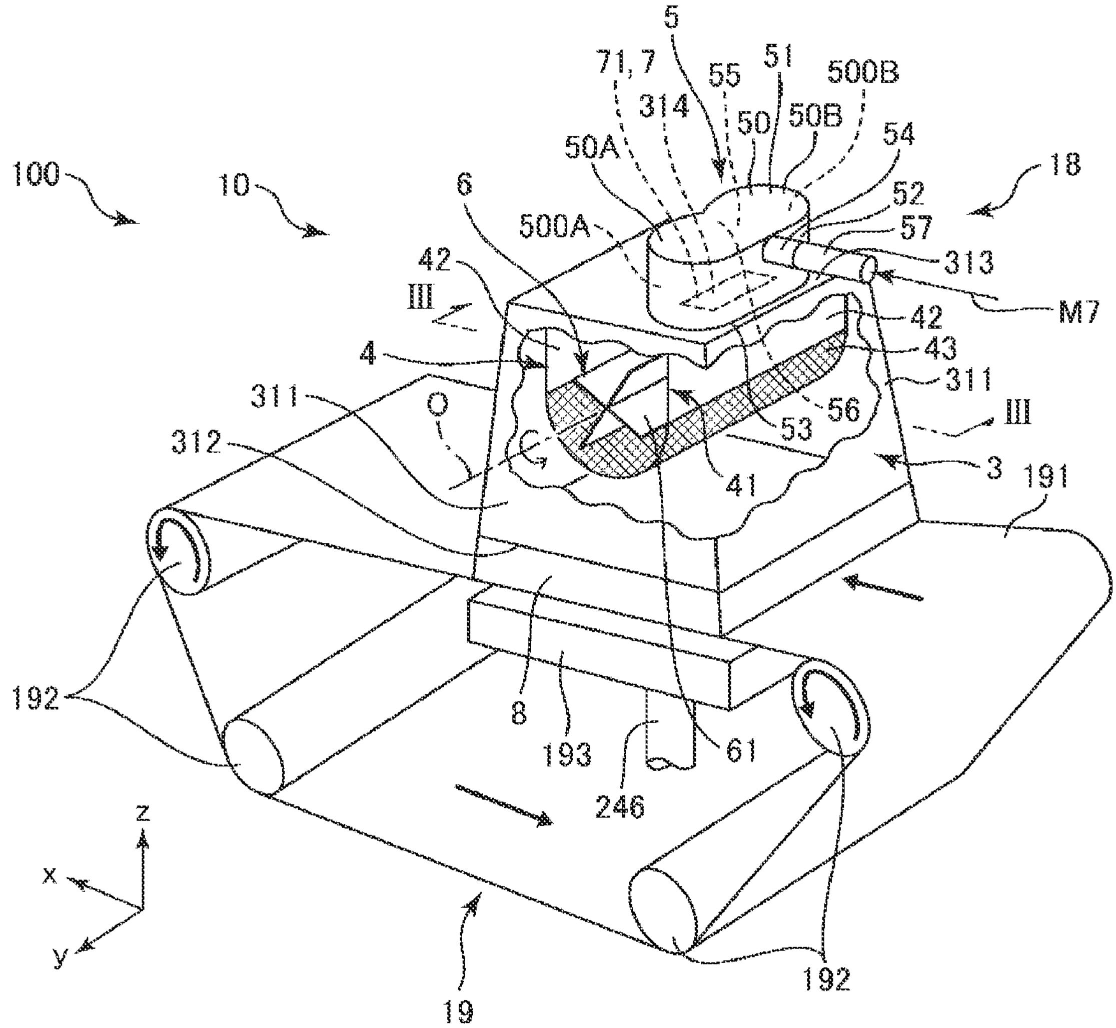
204
21
211
212
22
S
231
232
233
234
235
236
241
242
243
244
245
246
261
262
263
27
28
281
282
3
4
5
6
57
M1
M2
M3
M4-1
M4-2
M5
M6
M7
M8
x
y
z

FIG. 2


5
71, 7
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51
500B
314
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50A
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M7
III
42
4
43
311
311
O
III
53
56
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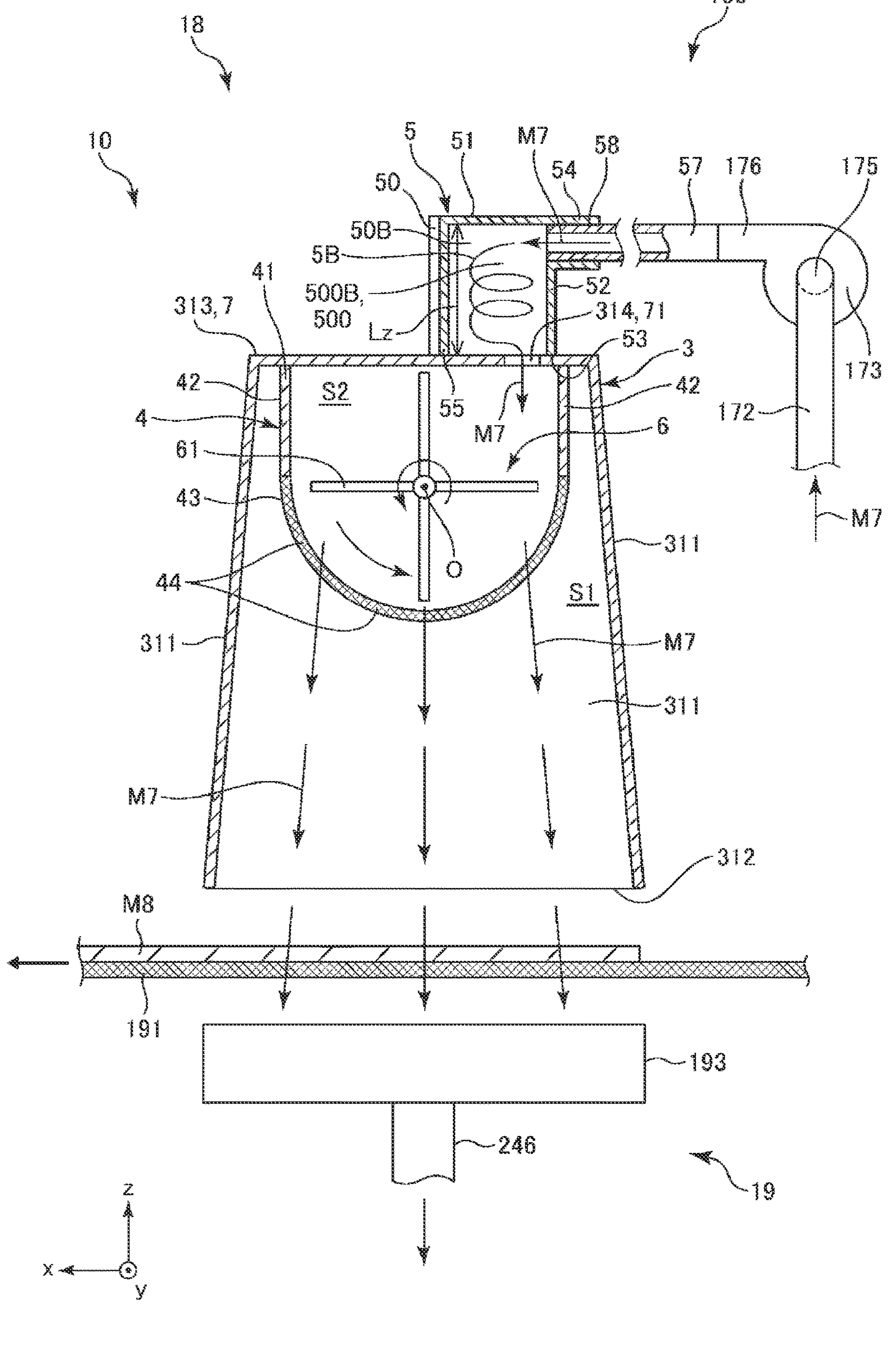
C
41
3
191
192
8
193
61
246
z
x
y
192
19

FIG. 3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3
42
42
S2
55
M7
6
4
61
43
O
44
S1
311
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M7
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M7
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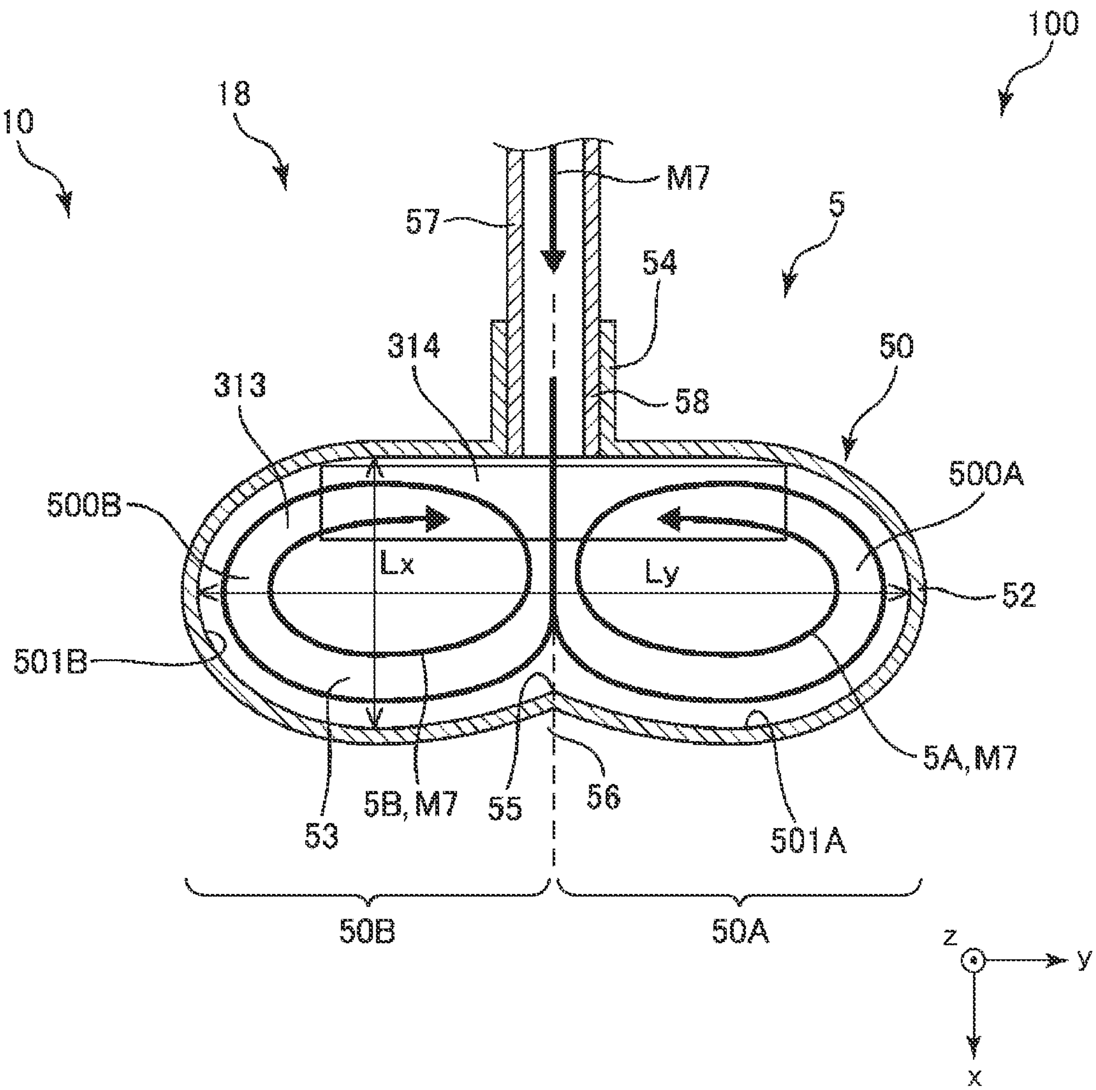
M8
191
193
246
19
173
172
M7
z
x
y

FIG. 4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10
18
M7
57
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313
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Lx
Ly
52
501B
5A, M7
53
5B, M7
55
56
501A
50B
50A
z
y
x

DISPERSION DEVICE AND ACCUMULATION DEVICE

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is based on, and claims priority from JP Application Serial Number 2022-210776, filed Dec. 27, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND OF THE INVENTION

(1) Field of the Invention

The present disclosure relates to a dispersion device and an accumulation device.

(2) Description of Related Art

In recent years, a dry-type sheet manufacturing apparatus that uses water as little as possible is proposed. As the dry-type sheet manufacturing apparatus, there is known a configuration including a defibrating section that defibrates a raw material containing fibers, such as waste paper, a dispersion section that disperses, in air, a defibrated material generated by the defibrating section, an accumulation section that accumulates the dispersed defibrated material, and a forming section that forms an accumulated material generated by the accumulation section into a sheet shape.

In a sheet manufacturing apparatus disclosed in JP-A-5-132843, the defibrated material is supplied to the dispersion section via a supply pipe, and the defibrated material is stirred and loosened in the dispersion section, and then dispersed.

However, in the apparatus disclosed in JP-A-5-132843, when a lump of the defibrated material that has not been sufficiently loosened is supplied to the dispersion section, the stirring in the dispersion section alone may not sufficiently loose the defibrated material depending on a size, an amount, or the like of the lump of the defibrated material. In this case, the defibrated material cannot be efficiently and satisfactorily dispersed, and there is a problem in that the dispersion section or the like is clogged because of the lump of the remaining defibrated material, which causes a decrease in processing efficiency, apparatus failure, apparatus stoppage, and the like.

BRIEF SUMMARY OF THE INVENTION

According to an aspect of the present disclosure, there is provided a dispersion device including: a supply section having a supply pipe for supplying a material containing fibers together with air, and a chamber that has a first swirl flow forming portion forming a first swirl flow of the air containing the material and a second swirl flow forming portion communicating with the first swirl flow forming portion and forming a second swirl flow in a direction opposite to the first swirl flow of the air containing the material and that is coupled to the supply pipe; a dispersion section that has a casing formed with a discharge port for discharging the material, stirs the material in the casing, and discharges and disperses the material from the discharge port into air; and a coupling section having a communication port through which the first swirl flow forming portion and the second swirl flow forming portion communicate with the casing.

According to another aspect of the present disclosure, there is provided an accumulation device including: a supply section having a supply pipe for supplying a material containing fibers together with air, and a chamber that has a first swirl flow forming portion forming a first swirl flow of the air containing the material and a second swirl flow forming portion communicating with the first swirl flow forming portion and forming a second swirl flow in a direction opposite to the first swirl flow of the air containing the material and that is coupled to the supply pipe; a dispersion section that has a casing formed with a discharge port for discharging the material, stirs the material in the casing, and discharges and disperses the material from the discharge port into air; a coupling section having a communication port through which the first swirl flow forming portion and the second swirl flow forming portion communicate with the casing; and an accumulation section accumulating the material dispersed by the dispersion section.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic side view illustrating a sheet manufacturing apparatus including a dispersion device and an accumulation device according to an embodiment of the present disclosure.

FIG. **2** is a perspective view of the dispersion device and the accumulation device shown in FIG. **1**.

FIG. **3** is a sectional view taken along the line III-III in FIG. **2**.

FIG. **4** is a cross-sectional plan view of a supply section illustrated in FIG. **2**.

DETAILED DESCRIPTION OF THE INVENTION

Hereinafter, a dispersion device and an accumulation device of the present disclosure will be described in detail based on preferred embodiments shown in the accompanying drawings.

Embodiment

FIG. **1** is a schematic side view illustrating a sheet manufacturing apparatus including a dispersion device and an accumulation device according to an embodiment of the present disclosure. FIG. **2** is a perspective view of the dispersion device and the accumulation device shown in FIG. **1**. FIG. **3** is a sectional view taken along the line III-III in FIG. **2**. FIG. **4** is a cross-sectional plan view of a supply section illustrated in FIG. **2**.

In the following, for convenience of description, as shown in FIGS. **1** to **4**, three axes orthogonal to each other are referred to as an x axis, a y axis, and a z axis. In addition, an xy plane including the x axis and the y axis is a horizontal plane, and the z axis is vertical. The state viewed from a z axis direction is referred to as "plan view". In addition, a direction in which an arrow of each axis points is referred to as "+", and the opposite direction is referred to as "−". In addition, an upper side of FIGS. **1**, **2**, and **3** is referred to as "upper" or "above", and a lower side thereof is referred to as "lower" or "below". In addition, in each drawing, a tip in a direction in which a material containing fibers flows is referred to as "downstream", and the opposite side is referred to as "upstream".

As illustrated in FIG. **1**, a sheet manufacturing apparatus **100** includes an accumulation device **10** that is an example of an accumulation device of the present disclosure, a sheet forming section **20**, a cutting section **21**, a stock section **22**, and a collection section **27**. The accumulation device **10** includes a raw material supply section **11**, a crushing section **12**, a defibrating section **13**, a sorting section **14**, a first web forming section **15**, a subdivision section **16**, a mixing section **17**, a dispersion device **18** that is an example of a dispersion device of the present disclosure, a second web forming section **19**, and a controller **28**.

In addition, the sheet manufacturing apparatus **100** includes a humidification section **231**, a humidification section **232**, a humidification section **233**, a humidification section **234**, a humidification section **235**, and a humidification section **236**. In addition, the sheet manufacturing apparatus **100** includes a blower **173**, a blower **261**, a blower **262**, and a blower **263**.

In the sheet manufacturing apparatus **100**, a raw material supply process, a crushing process, a defibrating process, a sorting process, a first web forming process, a fragmenting process, a mixing process, a dispersing process, a second web forming process, a sheet forming process, and a cutting process are executed in this order.

Hereinafter, a configuration of each section will be described.

As illustrated in FIG. **1**, the raw material supply section **11** is a portion that performs a raw material supply process of supplying a raw material M**1** to the crushing section **12**. As the raw material M**1**, a sheet-like material formed of a fiber-containing material containing cellulose fibers can be used. The cellulose fibers need only be a fibrous material mainly composed of cellulose as a compound, and may contain hemicellulose and lignin in addition to the cellulose. In addition, the raw material M**1** may be in any form, such as woven fabric or non-woven fabric. In addition, the raw material M**1** may be, for example, recycled paper manufactured by defibrating waste paper or YUPO paper (registered trademark) that is synthetic paper, or need not be recycled paper. In the present embodiment, the raw material M**1** is used or unnecessary waste paper.

The crushing section **12** is a portion that performs a crushing process of crushing, in the air such as in the atmosphere, the raw material M**1** supplied from the raw material supply section **11**. The crushing section **12** has a pair of crushing blades **121** and a chute **122**.

By rotating the pair of crushing blades **121** in opposite directions, the raw material M**1** can be crushed therebetween, that is, cut into crushed pieces M**2**. The shape and size of the crushed pieces M**2** are preferably suitable for a defibrating process in the defibrating section **13**. For example, the crushed pieces M**2** are preferably small pieces with a side length of 100 mm or less, and more preferably small pieces with a side length of 10 mm or more and 70 mm or less.

The chute **122** is disposed below the pair of crushing blades **121** and has, for example, a funnel shape. Thereby, the chute **122** can receive the crushed pieces M**2** that falls by being crushed by the crushing blades **121**.

In addition, above the chute **122**, the humidification section **231** is disposed adjacent to the pair of crushing blades **121**. The humidification section **231** humidifies the crushed pieces M**2** in the chute **122**. The humidification section **231** is configured of a vaporization type humidifier, particularly a warm air vaporization type humidifier, which has a filter (not illustrated) containing moisture and supplies humidified air with increased humidity to the crushed pieces M**2** by passing air through the filter. By supplying humidified air to the crushed pieces M**2**, it is possible to suppress adhesion of the crushed pieces M**2** to the chute **122** or the like due to static electricity.

The chute **122** is coupled to the defibrating section **13** via a pipe **241**. The crushed pieces M**2** collected in the chute **122** pass through the pipe **241** and are transported to the defibrating section **13**.

The defibrating section **13** is a portion that performs a defibrating process of defibrating the crushed pieces M**2** in the air, that is, in a dry manner. By performing the defibrating process in the defibrating section **13**, a defibrated material M**3** can be generated from the crushed pieces M**2**. Here, the term "defibrating" means unraveling the crushed pieces M**2** formed by binding a plurality of fibers, into individual fibers. Then, the unraveled material becomes the defibrated material M**3**. The shape of the defibrated material M**3** is a linear shape or a belt shape. In addition, the defibrated materials M**3** may exist in a state of being intertwined into a mass, that is, in a state of forming a so-called "lump".

For example, in the present embodiment, the defibrating section **13** includes an impeller having a rotor that rotates at a high speed and a liner that is located on an outer periphery of the rotor. The crushed pieces M**2** that flowed into the defibrating section **13** are defibrated by being interposed between the rotor and the liner.

In addition, the defibrating section **13** can generate a flow of air from the crushing section **12** toward the sorting section **14**, that is, an airflow, by the rotation of the rotor. Thereby, the crushed pieces M**2** can be sucked into the defibrating section **13** from the pipe **241**. In addition, after the defibrating process, the defibrated material M**3** can be sent to the sorting section **14** via a pipe **242**.

The blower **261** is installed in the middle of the pipe **242**. The blower **261** is an airflow generation device that generates an airflow toward the sorting section **14**. This facilitates the sending of the defibrated material M**3** to the sorting section **14**.

The sorting section **14** is a portion that performs a sorting process of sorting the defibrated material M**3** according to the length of the fibers. In the sorting section **14**, the defibrated material M**3** is sorted into a first sorted material M**4-1** and a second sorted material M**4-2**, which is larger than the first sorted material M**4-1**. The first sorted material M**4-1** has a size suitable for the subsequent manufacture of a sheet S. The average length thereof is preferably 1 μm or more and 30 μm or less. On the other hand, the second sorted material M**4-2** includes, for example, those with insufficient defibration and those in which the defibrinated fibers are excessively aggregated.

The sorting section **14** has a drum portion **141** and a housing portion **142** that houses the drum portion **141**.

The drum portion **141** is formed of a cylindrical net body, and is a sieve that rotates around a central axis thereof. The defibrated material M**3** flows into the drum portion **141**. Then, as the drum portion **141** rotates, the defibrated material M**3** smaller than a mesh opening of the net is sorted as the first sorted material M**4-1**, and the defibrated material M**3** having a size equal to or larger than the mesh opening of the net is sorted as the second sorted material M**4-2**.

The first sorted material M**4-1** falls from the drum portion **141**.

On the other hand, the second sorted material M**4-2** is sent to a pipe **243** coupled to the drum portion **141**. A part of the pipe **243** on a side opposite to the drum portion **141**, that is, an upstream part of the pipe **243** is coupled to the pipe **241**. The second sorted material M**4-2** that passed through the pipe **243** joins the crushed pieces M**2** in the pipe **241** and flows into the defibrating section **13** together with the crushed pieces M**2**. Thereby, the second sorted material M**4-2** is returned to the defibrating section **13**, and is defibrated together with the crushed pieces M**2**.

In addition, the first sorted material M**4-1** falls from the drum portion **141** while being dispersed in the air, and travels to the first web forming section **15** located below the drum portion **141**. The first web forming section **15** is a portion that performs a first web forming process of forming a first web M**5** from the first sorted material M**4-1**. The first web forming section **15** has a mesh belt **151**, three tension rollers **152**, and a suction portion **153**.

The mesh belt **151** is an endless belt, and the first sorted material M**4-1** is accumulated thereon. The mesh belt **151** is hung around the three tension rollers **152**. Then, the first sorted material M**4-1** on the mesh belt **151** is transported to the downstream by the rotational drive of the tension rollers **152**.

The first sorted material M**4-1** has a size equal to or larger than a mesh opening of the mesh belt **151**. Thereby, the first sorted material M**4-1** is restricted from passing through the mesh belt **151**, and therefore can be accumulated on the mesh belt **151**. In addition, the first sorted material M**4-1** is transported to the downstream together with the mesh belt **151** while being accumulated on the mesh belt **151**, so that the first sorted material M**4-1** is formed as a layered first web M**5**.

In addition, there is a concern that dust, dirt, or the like is mixed in the first sorted material M**4-1**. Dust or dirt may be generated by, for example, crushing or defibrating. Then, such dust or dirt is collected in the collection section **27**, which will be described below.

The suction portion **153** is a suction mechanism that sucks air from below the mesh belt **151**. Thereby, dust or dirt that passed through the mesh belt **151** can be sucked together with air.

In addition, the suction portion **153** is coupled to the collection section **27** via a pipe **244**. The dust or dirt sucked by the suction portion **153** is collected in the collection section **27**.

A pipe **245** is further coupled to the collection section **27**. In addition, the blower **262** is installed in the middle of the pipe **245**. By operating the blower **262**, a suction force can be generated in the suction portion **153**. This facilitates the formation of the first web M**5** on the mesh belt **151**. This first web M**5** is free of the dust or dirt. In addition, the dust or dirt passes through the pipe **244** and reaches the collection section **27** by the operation of the blower **262**.

The housing portion **142** is coupled to the humidification section **232**. The humidification section **232** is configured of a vaporization type humidifier similar to the humidification section **231**. Thereby, humidified air is supplied into the housing portion **142**. The first sorted material M**4-1** can be humidified with the humidified air, thereby also suppressing adhesion of the first sorted material M**4-1** to an inner wall of the housing portion **142** due to electrostatic force.

The humidification section **235** is disposed downstream of the sorting section **14**. The humidification section **235** is configured of an ultrasonic humidifier that sprays water. Thereby, moisture can be supplied to the first web M**5**, and thus the amount of moisture of the first web M**5** is adjusted. By this adjustment, the adsorption of the first web M**5** to the mesh belt **151** due to electrostatic force can be suppressed. Thereby, the first web M**5** is easily peeled off from the mesh belt **151** at a position where the mesh belt **151** is folded back by the tension rollers **152**.

The subdivision section **16** is disposed downstream of the humidification section **235**. The subdivision section **16** is a portion that performs a fragmenting process of fragmenting the first web M**5** peeled off from the mesh belt **151**. The subdivision section **16** has a propeller **161** that is supported rotatably, and a housing portion **162** that houses the propeller **161**. Then, the first web M**5** can be fragmented by the rotating propeller **161**. The fragmented first webs M**5** become subdivided bodies M**6**. In addition, the subdivided bodies M**6** descend in the housing portion **162**.

The housing portion **162** is coupled to the humidification section **233**. The humidification section **233** is configured of a vaporization type humidifier similar to the humidification section **231**. Thereby, humidified air is supplied into the housing portion **162**. The humidified air can also suppress adhesion of the subdivided bodies M**6** to the propeller **161** or an inner wall of the housing portion **162** due to electrostatic force.

The mixing section **17** is disposed downstream of the subdivision section **16**. The mixing section **17** is a portion that performs a mixing process of mixing the subdivided bodies M**6** and a binder P**1**. The mixing section **17** has a binder supply portion **171**, a pipe **172**, and a blower **173**.

An upstream end part of the pipe **172** is coupled to the housing portion **162** of the subdivision section **16**, and a downstream end part of the pipe **172** is coupled to a suction port **175** of the blower **173**. By operating the blower **173**, a mixture M**7** of the subdivided bodies M**6** and the binder P**1** is sent toward a downstream part in the pipe **172**.

The binder supply portion **171** is coupled in the middle of the pipe **172**. The binder supply portion **171** has a screw feeder **174**. When the screw feeder **174** is rotationally driven, the binder P**1** can be quantitatively supplied to the pipe **172** as powders or particles. The binder P**1** supplied to the pipe **172** is mixed with the subdivided bodies M**6** at a desired ratio to form the mixture M**7**.

Examples of the binder P**1** include: natural product-derived ingredients such as starch, dextrin, glycogen, amylose, hyaluronic acid, arrowroot, konjac, potato starch, etherified starch, esterified starch, natural gum glue, fiber-derived glue, seaweed, and animal protein; polyvinyl alcohol; polyacrylic acid; and polyacrylamide, and one or two or more selected from these can be used in combination. However, a natural product-derived ingredient is preferable, and starch is more preferable. In addition, for example, thermoplastic resins such as various polyolefins, acrylic resins, polyvinyl chloride, polyesters, and polyamides; and various thermoplastic elastomers can be used.

In addition to the binder P**1**, the material supplied from the binder supply portion **171** may include, for example, a colorant for coloring fibers, an aggregation suppressing agent for suppressing aggregation of fibers or aggregation of the binder P**1**, a flame retardant for making fibers and the like less flammable, and a paper strength enhancer for enhancing a paper strength of the sheet S. Alternatively, the materials are contained and compounded in the binder P**1** beforehand, and the resultant may be supplied from the binder supply portion **171**.

The blower **173** is installed downstream of the pipe **172**, the dispersion device **18** is installed downstream of the blower **173**, and the second web forming section **19** is installed downstream of the dispersion device **18**. An upstream end part of a supply pipe **57** of the dispersion device **18** is coupled to an ejection port **176** of the blower **173**.

The subdivided bodies M**6** and the binder P**1** in the pipe **172** are introduced into the blower **173** by an airflow generated by the action of a rotating blade installed inside the blower **173**, and are stirred and mixed. In addition, the blower **173** discharges the airflow toward the downstream from the ejection port by the action of the rotating blade. That is, an airflow toward the dispersion device **18** is generated. Such an airflow enables the stirring and mixing of the subdivided bodies M**6** and the binder P**1**, and the resulting mixture M**7** flows through the supply pipe **57** into the dispersion device **18** in a state where the subdivided bodies M**6** and the binder P**1** are uniformly dispersed. In addition, the subdivided bodies M**6** in the mixture M**7** are loosened in the process of passing through the pipe **172** and the blower **173** to have a finer fibrous shape.

The dispersion device **18** performs a dispersing process of loosening intertwined fibers in a material containing fibers, that is, in the mixture M**7**, and dispersing the fibers in the air. A configuration of the dispersion device **18** will be described in detail below. The mixture M**7** dispersed in the air by the dispersion device **18** falls, and travels to the second web forming section **19** located below the dispersion device **18**.

The second web forming section **19** is an accumulation section that accumulates the mixture M**7** dispersed by the dispersion device **18**, and is a portion that performs a second web forming process of forming a second web M**8** from the mixture M**7**. The second web forming section **19** has a mesh belt **191**, four tension rollers **192**, and a suction portion **193**.

The mesh belt **191** is an endless belt, and the mixture M**7** is accumulated thereon. The mesh belt **191** is hung around the four tension rollers **192**. Then, the mixture M**7** on the mesh belt **191** is transported to the downstream by the rotational drive of the tension rollers **192**.

In addition, most of the mixture M**7** on the mesh belt **191** has a size equal to or larger than a mesh opening of the mesh belt **191**. Thereby, the mixture M**7** is restricted from passing through the mesh belt **191**, and therefore can be accumulated on the mesh belt **191**. In addition, the mixture M**7** is transported to the downstream together with the mesh belt **191** while being accumulated on the mesh belt **191**, so that the mixture M**7** is formed as a layered second web M**8**.

The suction portion **193** is a suction mechanism that sucks air from below the mesh belt **191**. Thereby, the mixture M**7** can be sucked onto the mesh belt **191**, and thus, this facilitates the accumulation of the mixture M**7** on the mesh belt **191**.

A pipe **246** is coupled to the suction portion **193**. In addition, the blower **263** is installed in the middle of the pipe **246**. By operating the blower **263**, a suction force can be generated in the suction portion **193**.

The humidification section **236** is disposed downstream of the dispersion device **18**. The humidification section **236** is configured of an ultrasonic humidifier similar to the humidification section **235**. Thereby, moisture can be supplied to the second web M**8**, and thus the amount of moisture of the second web M**8** is adjusted. By this adjustment, the adsorption of the second web M**8** to the mesh belt **191** due to electrostatic force can be suppressed. Thereby, the second web M**8** is easily peeled off from the mesh belt **191** at a position where the mesh belt **191** is folded back by the tension rollers **192**.

The total amount of moisture added to the humidification section **231** to the humidification section **236** is, for example, preferably 0.5 parts by mass or more and 20 parts by mass or less with respect to 100 parts by mass of the material before humidification.

The sheet forming section **20** is disposed downstream of the second web forming section **19**. The sheet forming section **20** is a portion that performs a sheet forming process of forming the sheet S from the second web M**8**. The sheet forming section **20** has a pressurizing portion **201** and a heating portion **202**.

The pressurizing portion **201** has a pair of calender rollers **203**, and can pressurize the second web M**8** between the calender rollers **203** without heating the second web M**8**. Thereby, a density of the second web M**8** is increased. An extent of the heating at this time is preferably, for example, such that the binder P**1** is not melted. Then, the second web M**8** is transported toward the heating portion **202**. One of the pair of calender rollers **203** is a main roller driven by an operation of a motor (not illustrated), and the other is a driven roller.

The heating portion **202** has a pair of heating rollers **204**, and can pressurize the second web M**8** while heating the second web M**8** between the heating rollers **204**. By this heating and pressurization, the binder P**1** is melted in the second web M**8**, and fibers are bound to each other through the melted binder P**1**. Thereby, the sheet S is formed. The sheet S is transported toward the cutting section **21**. One of the pair of heating rollers **204** is a main roller driven by an operation of a motor (not illustrated), and the other is a driven roller.

The cutting section **21** is disposed downstream of the sheet forming section **20**. The cutting section **21** is a portion that performs a cutting process of cutting the sheet S. The cutting section **21** has a first cutter **211** and a second cutter **212**.

The first cutter **211** cuts the sheet S in a direction intersecting a transport direction of the sheet S, particularly in a direction orthogonal to the transport direction.

The second cutter **212** is located downstream of the first cutter **211**, and cuts the sheet S in a direction parallel to the transport direction of the sheet S. The cutting is a process of removing unnecessary portions at both end parts of the sheet S, that is, end parts in the +y axis direction and in the −y axis direction to adjust a width of the sheet S. In addition, the portion removed by the cutting is referred to as a so-called "offcut".

Through such cutting with the first cutter **211** and the second cutter **212**, the sheet S having a desired shape and size can be obtained. The sheet S is transported further downstream and accumulated in the stock section **22**.

Each section included in such a sheet manufacturing apparatus **100** is electrically coupled to the controller **28**. The operations of these sections are controlled by the controller **28**.

The controller **28** has a central processing unit (CPU) **281** and a storage **282**. For example, the CPU **281** can make various determinations and various commands.

The storage **282** stores various programs, such as a program for manufacturing the sheet S, various calibration curves, a table, and the like.

The controller **28** may be built in the sheet manufacturing apparatus **100** or may be provided in an external device such as an external computer. For example, the external device may communicate with the sheet manufacturing apparatus **100** via a cable or the like, may wirelessly communicate with the sheet manufacturing apparatus **100**, or may be connected to the sheet manufacturing apparatus **100** via a network such as the Internet.

In addition, for example, the CPU **281** and the storage **282** may be integrated into one unit, the CPU **281** may be built in the sheet manufacturing apparatus **100** and the storage **282** may be provided in an external device such as an external computer, or the storage **282** may be built in the sheet manufacturing apparatus **100** and the CPU **281** may be provided in an external device such as an external computer.

Next, the dispersion device **18** will be described.

As illustrated in FIGS. **2** and **3**, the dispersion device **18** includes a dispersion section **4** that disperses the mixture M7, a supply section **5** that supplies the mixture M7 to the dispersion section **4**, a coupling section **7** that couples the supply section **5** and the dispersion section **4**, and a housing **3** that covers the dispersion section **4**.

The housing **3** is configured of a casing having four side walls **311** and a top plate **313** located above the side walls **311**. A space S1 surrounded by the four side walls **311** and the top plate **313** is formed inside the housing **3**, and the dispersion section **4** is housed in the space S**1**. Therefore, the space S1 is also referred to as a dispersion space. In addition, most of a portion between the dispersion section **4** and the mesh belt **191** is covered with the housing **3**.

The housing **3** has a lower opening **312** facing the mesh belt **191**. The lower opening **312** constitutes a discharge section that discharges the mixture M7, which is dispersed by the dispersion section **4** and descends in the space S1, toward the second web forming section **19**. A separation distance between the lower opening **312** and the mesh belt **191** is set to a value suitable for forming the second web M**8**, and is, for example, 0 mm or more and 10 mm or less.

At least one of the four side walls **311** constituting the housing **3** is inclined in a vertical direction. In the present embodiment, each of the four side walls **311** is inclined in the vertical direction, and forms a skirt portion that widens toward the lower opening **312**. In other words, the space S1 of the housing **3** has a shape in which an area of a cross section parallel to a horizontal plane gradually increases downward, that is, in the −z axis direction. Thereby, the stirring and loosening effects of the mixture M7 that descends in the space S1 toward the second web forming section **19** are more satisfactorily exhibited, and the second web M**8** with a desired area and thickness, that is, with a necessary and sufficient area and thickness can be formed on the mesh belt **191**.

The space S1 of the housing **3** may have a shape in which the area of the cross section parallel to the horizontal plane is constant along the z axis direction.

The mixture M7 is sufficiently stirred and loosened by the supply section **5** and the dispersion section **4**, and the loosening by stirring is continued in the space S1 of the housing **3**, so that a homogeneous and uniform accumulated material of the mixture M7 without a lump of fibers, that is, the second web M**8** is obtained in the second web forming section **19**.

The top plate **313** is provided with an opening **314**. The opening **314** is also a communication port **71** through which a stirring space **500** of the supply section **5** and an accommodation space S2 of the dispersion section **4** communicate with each other, and is configured of a long hole extending in the y axis direction, that is, in a first direction parallel to a rotation axis O. The mixture M7 supplied from the supply section **5** is supplied into the dispersion section **4** through the opening **314**.

In addition, as illustrated in FIG. **1**, the humidification section **234** is coupled to the housing **3**. The humidification section **234** is configured of a vaporization type humidifier similar to the humidification section **231**. Thereby, in the housing **3**, humidified air generated by the humidification section **234** is supplied to the space S1 in the housing **3**. The inside of the housing **3** can be humidified with the humidified air, thereby also suppressing adhesion of the mixture M7 dispersed by the dispersion section **4** to an inner wall of the housing **3** due to electrostatic force.

As illustrated in FIGS. **2** and **3**, the dispersion section **4** has a casing **41** and a stirring member **6** that rotates inside the casing **41**. The casing **41** is joined to a lower surface of the top plate **313** of the housing **3**, and has a pair of side walls **42** disposed parallel to each other and a porous screen **43** joined to lower ends of both side walls **42** and formed with the discharge port **44** for discharging the mixture M7. The discharge port **44** is configured of a plurality of small holes.

The pair of side walls **42** have an elongated shape extending in the y axis direction, and are disposed at a predetermined distance in the x axis direction with the opening **314** interposed therebetween.

The porous screen **43** has a semi-cylindrical shape extending in the y axis direction and curved and protruding downward, that is, in the −z axis direction. That is, the porous screen **43** has an arc shape at any position in the y axis direction when viewed in a cross section with the y axis as a normal line. Thereby, the mixture M7 can move smoothly in the dispersion section **4**, and the stirring is performed satisfactorily. In addition, two upper ends of the porous screen **43** are coupled to the lower ends of the pair of side walls **42**, respectively. An end part on the −y axis side and an end part on the +y axis side of the casing **41** are closed by shielding walls (not illustrated), respectively. A rotation axis of the stirring member **6** described below is supported so as to be rotatable by a pair of the shielding walls.

A space defined by the pair of side walls **42**, the porous screen **43**, the pair of shielding walls, and the top plate **313** is the accommodation space S2 in which the mixture M7 is accommodated. The accommodation space S2 has a function as a second stirring space for further stirring and loosening the mixture M7 with respect to the stirring space **500** of the supply section **5**, and the space S1 has a function as a third stirring space for further stirring and loosening the mixture M7 transferred from the accommodation space S2.

The porous screen **43** can be made of, for example, a net-like body such as a mesh or a plate material having a large number of through-holes. Thereby, the mixture M7 in the dispersion section **4** is discharged to an outside of the accommodation space S2 via the discharge port **44** of the porous screen **43** and dispersed into the space S1. In addition, by appropriately setting the size of a mesh opening or the size of the through-holes of the porous screen **43**, the mixture M7 having a desired fiber length can be preferentially dispersed and accumulated on the mesh belt **191**.

The stirring member **6** has a function of facilitating the dispersion of the mixture M7 from the porous screen **43** while stirring and loosening the mixture M7 supplied into the dispersion section **4** by rotating in the accommodation space S2 of the dispersion section **4**. The stirring member **6** has four blades **61** disposed around the rotation axis O at equal angular intervals. The blade **61** is made of an elongated plate material extending in the y axis direction. End parts on one long side of the blades **61** are coupled to each other, and the stirring member **6** rotates about the coupled portion as the center of rotation, that is, the rotation axis O. In the present embodiment, the stirring member **6** has a cross-shaped cross section with the rotation axis O as a normal line.

In addition, the stirring member **6** is coupled to a rotational drive source (not illustrated), and the operation of the rotational drive source is controlled by the controller **28** illustrated in FIG. **1**. In the present embodiment, the stirring member **6** rotates clockwise when viewed from the +y axis side.

By the rotation of the stirring member **6**, each blade **61** presses an appropriate amount of the mixture M7 against the porous screen **43** while stirring and loosening the mixture M7 in the accommodation space S2. Thereby, the mixture M7 can be evenly discharged and dispersed satisfactorily from the entire region of the porous screen **43** while preventing the mixture M7 from being excessively supplied and clogging the porous screen **43**.

In addition, the stirring member **6** rotates in a state where each blade **61** is separated from the side wall **42** and the porous screen **43**. Thereby, the rotation of the stirring member **6** can be smoothly performed, and the mixture M7 can be prevented from being pressurized excessively between the blade **61** and the porous screen **43**, so that more favorable dispersion can be performed.

In the present embodiment, a case where four blades **61** are provided is described, but the present disclosure is not limited to this, and for example, the number of the blades **61** may be one to three, or four or more. In addition, a case where each blade **61** has a flat plate shape is described, but the present disclosure is not limited to this, and for example, each blade **61** may have a shape curved in one direction when viewed in a cross section with the rotation axis O as a normal line. As described above, a configuration of the stirring member **6**, particularly the shape, the number, the disposition, and the like of the blades **61** are not limited to the illustrated configuration. In addition, in the dispersion section **4**, the stirring member **6** itself may be omitted, or a stirring mechanism different from the illustrated mechanism, for example, a mechanism having a stirring member that does not rotate but reciprocates may be installed.

The supply section **5** is installed above the top plate **313** of the housing **3**. As illustrated in FIG. **4**, the supply section **5** supplies, to the dispersion section **4**, the mixture M7 supplied from the supply pipe **57** while stirring and loosening the mixture M7 by a first swirl flow **5**A and a second swirl flow **5**B. The supply section **5** includes the chamber **50** having the stirring space **500** inside. The chamber **50** has a top plate **51** and a side wall **52** erected downward from an edge portion of the top plate **51**, that is, in the −z axis direction. The top plate **51** has a shape like glasses in plan view. The side wall **52** is provided to surround a space of a lower portion of the top plate **51** over the entire circumference of the edge portion of the top plate **51**.

A coupling port **54** is provided at an upper portion of the side wall **52**, that is, at a portion on the +z axis side and on the −x axis side. The coupling port **54** is a tubular port formed to protrude in the −x axis direction. An end part **58**, which is a downstream part, of the supply pipe **57** is coupled to the coupling port **54**. On the other hand, an end part, which is an upstream part, of the supply pipe **57** is coupled to the ejection port **176** of the blower **173**. By operating the blower **173**, the mixture M7 of the subdivided bodies M**6** and the binder P1 is ejected from the ejection port **176**, passes through the supply pipe **57** and the coupling port **54** sequentially, and flows into the chamber **50** together with air.

In the present embodiment, pipe axes of the end part **58** of the supply pipe **57** and the coupling port **54** are disposed parallel to the x axis direction. However, the present disclosure is not limited to this, and the end part **58** and the coupling port **54** may be disposed to be inclined at a predetermined angle with respect to the x axis.

In addition, a lower portion of the chamber **50** has a lower opening **53** that is open downward. The lower opening **53** is an opening formed along a lower end of the side wall **52**, that is, an end part on the −z axis side. The chamber **50** is joined to an upper surface of the top plate **313** such that the lower opening **53** is closed by the top plate **313** of the housing **3**.

The lower opening **53** includes the opening **314** when viewed in plan view, that is, when viewed in the z axis direction. Thereby, an inside of the chamber **50**, that is, a stirring space **500**A of a first swirl flow forming portion **50**A and a stirring space **500**B of a second swirl flow forming portion **50**B, and an inside of the casing **41**, that is, the accommodation space S2 communicate with each other via the lower opening **53** and the opening **314**. In other words, the opening **314** is the communication port **71** through which the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B communicate with the casing **41**.

The top plate **313** formed with the communication port **71** supports and fixes the casing **41** of the dispersion section **4** on a bottom surface side thereof, and supports and fixes the chamber **50** of the supply section **5** on an upper surface side thereof. That is, the casing **41** of the dispersion section **4** and the chamber **50** of the supply section **5** are coupled via the top plate **313**. Thereby, the top plate **313** functions as the coupling section **7** that couples the dispersion section **4** and the supply section **5**.

However, the present disclosure is not limited to this configuration, and the coupling section **7** may be configured of a coupling member such as a coupling pipe or a duct that couples the chamber **50** and the casing **41**, for example, with another configuration.

As illustrated in FIG. **4**, the chamber **50** has the first swirl flow forming portion **50**A that forms the first swirl flow **5**A of air containing the mixture M7, and the second swirl flow forming portion **50**B that communicates with the first swirl flow forming portion **50**A and that forms second swirl flow **5**B of air containing the mixture M7. A swirl direction of the first swirl flow **5**A is opposite to a swirl direction of the second swirl flow **5**B. The first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B communicate with each other via a boundary portion **56**.

The chamber **50** has the stirring space **500** for stirring and loosening the mixture M7 therein. The stirring space **500** is a space surrounded by the top plate **51**, the side wall **52**, and the top plate **313**. The stirring space **500** is configured of the stirring space **500**A and the stirring space **500**B that communicate with each other. An internal space of the first swirl flow forming portion **50**A is the stirring space **500**A, and an internal space of the second swirl flow forming portion **50**B is the stirring space **500**B.

The first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B are disposed side by side in the y axis direction, that is, in an extending direction of the opening **314**, or in an axial direction of the rotation axis O. The first swirl flow forming portion **50**A is located on the +y axis side, and the second swirl flow forming portion **50**B is located on the −y axis side. The end part **58** of the supply pipe **57** and the coupling port **54** are coupled to the boundary portion **56** between the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B.

A protrusion portion **55** is provided on a portion, on the +x axis side of the boundary portion **56**, of an inner surface of the side wall **52**, that is, a surface facing the stirring space **500**. The protrusion portion **55** is formed to protrude in a chevron shape toward the −x axis side, that is, toward the coupling port **54** side. The protrusion portion **55** has a width that narrows toward the −x axis, and has a sharp tip. The protrusion portion **55** is formed over the entire region in z axis direction. Even when the protrusion portion **55** is omitted, the effect of the present disclosure can be obtained.

The first swirl flow forming portion **50**A is a portion where the first swirl flow **5**A of air containing the mixture **M7** is formed, and the second swirl flow forming portion **50**B is a portion where the second swirl flow **5**B of air containing the mixture M7 is formed.

As illustrated in FIG. **4**, the inner surface of the side wall **52** of the first swirl flow forming portion **50**A is a first curved surface **501**A that is curved to protrude outward. In the first curved surface **501**A, a curvature of a portion on the +y axis side is larger than that of a portion on the +x axis side.

It is preferable that R**2**≥R**1**, and more preferable that R**2**>R**1**, in which a radius of curvature of the portion of the first curved surface **501**A on the +y axis side is R**1**, and a radius of curvature of the portion of the first curved surface **501**A on the +x axis side is R**2**. In this case, a value of R**1**/R**2** is not particularly limited, but is preferably 0.2 or more and 0.9 or less, and more preferably 0.3 or more and 0.75 or less. Thereby, a swirl flow more suitable for stirring can be formed.

The inner surface of the side wall **52** of the second swirl flow forming portion **50**B is a second curved surface **501**B that is curved to protrude outward. In the second curved surface **501**B, a curvature of a portion on the −y axis side is larger than that of a portion on the +x axis side. The magnitude relationships and ratios of radii of curvature of these portions are the same as those of the first curved surface **501**A.

As illustrated in FIG. **4**, the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B have a shape that is symmetrical with respect to the boundary portion **56** therebetween. That is, the first curved surface **501**A and the second curved surface **501**B have a shape that is symmetrical with respect to the boundary portion **56**. Thereby, the shapes of the first swirl flow **5**A and the second swirl flow **5**B can be formed in a well-balanced manner, and the strength and the swirl speed of both swirl flows can be made more uniform. The boundary portion **56** is configured of a plane parallel to the x-z plane.

Air containing the mixture M7 (hereinafter, simply referred to as "air") flowing through the supply pipe **57** in the downstream direction and supplied from the coupling port **54** to the stirring space **500** first advances in the +x axis direction in the stirring space **500**, and hits the protrusion portion **55** and is divided into the +y axis side and the −y axis side. That is, the air supplied from the coupling port **54** to the stirring space **500** is divided into the stirring space **500**A and the stirring space **500**B by the protrusion portion **55**.

Here, it is preferable that the amount of the air that is divided and flows into the stirring space **500**A, that is, the amount of the mixture M7 is substantially equal to the amount of the air that flows into the stirring space **500**B, that is, the amount of the mixture M7, but the present disclosure is not limited to this, and for example, a ratio of the former air amount VA to the latter air amount VB may be in a range of 1:5 to 5:1.

The air divided into the stirring space **500**A flows downward (in the −z axis direction) and toward a center portion of the swirling while swirling counterclockwise in FIG. **4** along the first curved surface **501**A, to form the first swirl flow **5**A. On the other hand, the air divided into the stirring space **500**B flows downward (in the −z axis direction) and toward a center portion of the swirling while swirling clockwise in FIG. **4** along the second curved surface **501**B, to form the second swirl flow **5**B as illustrated in FIG. **3**. When the first swirl flow **5**A and the second swirl flow **5**B reach a lower portion of the stirring space **500**, the first swirl flow **5**A and the second swirl flow **5**B travel toward the opening **314** formed in the top plate **313**, that is, the communication port **71**.

The first swirl flow **5**A and the second swirl flow **5**B are airflows that travel toward the opening **314** while swirling in opposite directions. The mixture M7 supplied from the coupling port **54** together with air is divided in the vicinity of the protrusion portion **55**, and is stirred and loosened with the airflow of each of the first swirl flow **5**A and the second swirl flow **5**B. Then, the first swirl flow **5**A and the second swirl flow **5**B containing the mixture M7 join together in the vicinity of the opening **314**, and pass through the opening **314** and flow into the dispersion section **4** in a state where the stirring is further promoted and the mixture M7 is sufficiently loosened.

As described above, the supply section **5** supplies the mixture M7 to the dispersion section **4** in a state where the mixture M7 is stirred and loosened by the first swirl flow **5**A and the second swirl flow **5**B, prior to dispersion of the mixture M7 by the dispersion section **4**. Thereby, the dispersion section **4** can more satisfactorily disperse the mixture M7. That is, when the mixture M7 passes through the discharge port **44** of the porous screen **43**, the mixture M7 can be evenly dispersed from the entire region of the porous screen **43** while preventing the discharge port **44** from being clogged. Thereby, the mixture M7 can be smoothly and satisfactorily dispersed.

As illustrated in FIGS. **3** and **4**, when the length (maximum length) of the stirring space **500** in the x axis direction is Lx, the length (maximum length) of the stirring space **500** in the y axis direction is Ly, and the length (maximum length) of the stirring space **500** in the z axis direction is Lz, it is preferable that the following relationship is satisfied.

Ly/Lx is not particularly limited, but is preferably 1.0 or more and 5.0 or less, and more preferably 2.0 or more and 4.0 or less. Thereby, the first swirl flow **5**A and the second swirl flow **5**B can be formed more satisfactorily, and the stirring and loosening effects of the mixture M7 are enhanced.

Lz/Lx is not particularly limited, but is preferably 0.5 or more and 10.0 or less, and more preferably 1.0 or more and 5.0 or less. Thereby, the length of the stirring space **500** in the z axis direction, that is, the pass length of the first swirl flow **5**A and the second swirl flow **5**B can be sufficiently ensured, and the mixture M7 can be sufficiently stirred and loosened.

Although not illustrated, a straightening plate can also be provided inside the chamber **50**. Thereby, the shapes of the first swirl flow **5**A and the second swirl flow **5**B can be formed more satisfactorily, and the loosening effect of the mixture M7 by the stirring can be further enhanced.

As illustrated in FIG. **3**, the opening **314** is provided at a position that does not overlap the rotation axis O when viewed in plan view, that is, when viewed in the z axis direction. That is, the opening **314** is provided on the −x axis side with respect to the rotation axis O. Thereby, the mixture M7 supplied from the supply section **5** to the dispersion section **4** immediately collides with the blade **61** of the stirring member **6** that rotates directly below the opening **314**. Accordingly, the stirring by the stirring member **6** can be performed more satisfactorily. In particular, as illustrated in FIG. **3**, when the opening **314** is provided on the −x axis side with respect to the rotation axis O, and the stirring member **6** rotates counterclockwise when viewed from the +y axis side, the mixture M7 that passes through the opening **314** and travels downward collides head-on with the rising blade **61**. Accordingly, the stirring by the stirring member **6** can be performed more efficiently and satisfactorily, and the loosening effect of the mixture M7 is further enhanced.

The present disclosure is not limited to the above configuration, and the opening **314** may be provided on the +x axis side with respect to the rotation axis O in plan view, or may be provided at a position overlapping the rotation axis O in plan view. When the opening **314** is provided on the +x axis side with respect to the rotation axis O, for example, even when the fiber length of the fibers of the mixture M7 is relatively long or the amount of the mixture M7 supplied per unit time is large, there is an advantage that it is difficult to form a lump in the dispersion section **4**.

In addition, the stirring member **6** may be configured such that the rotation direction thereof can be switched between the clockwise rotation and the counterclockwise rotation. In this case, when the opening **314** is provided at a position that does not overlap the rotation axis O in plan view, any of the above-described effects can be selectively obtained by switching the rotation direction of the stirring member **6**.

As described above, the dispersion device **18** includes the supply section **5** having the supply pipe **57** for supplying the mixture M7, which is a material containing fibers, together with air, and the chamber **50** that has the first swirl flow forming portion **50**A forming the first swirl flow **5**A of the air containing the mixture M7 and the second swirl flow forming portion **50**B communicating with the first swirl flow forming portion **50**A and forming the second swirl flow **5**B in a direction opposite to the first swirl flow **5**A of the air containing the mixture M7 and that is coupled to the supply pipe **57**, the dispersion section **4** that has the casing **41** formed with the discharge port **44** for discharging the mixture M7, stirs the mixture M7 in the casing **41**, and discharges and disperses the mixture M7 from the discharge port **44** into air, and the coupling section **7** having the communication port **71** through which the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B communicate with the casing **41**. Thereby, the supply section **5** can supply the mixture M7 to the dispersion section **4** in a state where the mixture M7 is sufficiently stirred and loosened by the first swirl flow **5**A and the second swirl flow **5**B in the opposite directions, prior to dispersion of the mixture M7 by the dispersion section **4**. Accordingly, the dispersion section **4** can smoothly and satisfactorily disperse the mixture M7 without causing clogging or the like in the discharge port **44**.

In addition, the accumulation device **10** includes the supply section **5** having the supply pipe **57** for supplying the mixture M7, which is a material containing fibers, together with air, and the chamber **50** that has the first swirl flow forming portion **50**A forming the first swirl flow **5**A of the air containing the mixture M7 and the second swirl flow forming portion **50**B communicating with the first swirl flow forming portion **50**A and forming the second swirl flow **5**B in a direction opposite to the first swirl flow **5**A of the air containing the mixture M7 and that is coupled to the pipe **172**, the dispersion section **4** that has the casing **41** formed with the discharge port **44** for discharging the mixture M7, stirs the mixture M7 in the casing **41**, and discharges and disperses the mixture M7 from the discharge port **44** into air, the coupling section **7** having the communication port **71** through which the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B communicate with the casing **41**, and the second web forming section **19** serving as an accumulation section accumulating the mixture M7 dispersed by the dispersion section **4**. Thereby, the supply section **5** can supply the mixture M7 to the dispersion section **4** in a state where the mixture M7 is sufficiently stirred and loosened by the first swirl flow **5**A and the second swirl flow **5**B in the opposite directions, prior to dispersion of the mixture M7 by the dispersion section **4**. Accordingly, the dispersion section **4** can smoothly and satisfactorily disperse the mixture M7 without causing clogging or the like in the discharge port **44**. As a result, in the accumulation section, the second web M**8** that is a favorable accumulated material having a uniform thickness can be obtained. In addition, in the accumulation section, a homogeneous and uniform second web M**8** without the lump of fibers is obtained.

In addition, as described above, the dispersion section **4** has the stirring member **6** installed in the casing **41** and rotating around the rotation axis O. Thereby, the mixture M7 stirred and loosened in the supply section **5** can be further stirred and loosened by the stirring member **6**. Accordingly, the dispersion section **4** can further smoothly and satisfactorily disperse the mixture M7 due to the synergistic effect of these two-stage loosening.

The stirring member **6** may be omitted. In this case, for example, it is preferable to form an airflow in the casing **41** to stir and loosen the mixture M7.

In addition, as described above, the communication port **71** has an elongated shape extending along the first direction parallel to the rotation axis O. Thereby, the supply section **5** can supply the mixture M7 to the dispersion section **4** such that the mixture M7 is present at any position in the first direction. Accordingly, the mixture M7 can be stirred and loosened more evenly and satisfactorily by the stirring member **6**. As a result, the dispersion section **4** can more satisfactorily disperse the mixture M7.

The present disclosure is not limited to the above configuration, and the communication port **71** (opening **314**) may be configured of a plurality of holes, and the holes may be disposed side by side in the y axis direction, that is, in the first direction. In addition, the plurality of holes disposed in the y axis direction may be disposed in a plurality of rows in the x axis direction.

In addition, the coupling section **7** may have a configuration in which the shape, dimension, or opening area of the communication port **71** (opening **314**) can be adjusted. Examples of a method of adjusting the opening area of the communication port **71** include installing a shutter that shields the communication port **71** so that an opening degree of the communication port **71** can be changed continuously or stepwise. In addition, the coupling section **7** may have a configuration in which the formation position of the communication port **71** with respect to the supply section **5** and the dispersion section **4** can be adjusted. Thereby, the optimum condition of the communication port **71** for loosening the mixture M7 by the stirring can be set according to various conditions such as the supply amount, the flow velocity, and the flow rate of the mixture M7 from the supply pipe **57**.

In addition, as described above, the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B are disposed side by side in the first direction parallel to the rotation axis O. Thereby, the supply section **5** can supply the mixture M7 to the dispersion section **4** such that the sufficiently loosened mixture M7 is present at any position in the first direction. Accordingly, the mixture M7 can be stirred and loosened further evenly and satisfactorily by the stirring member **6**. As a result, the dispersion section **4** can more satisfactorily disperse the mixture M7.

In addition, as described above, an inner peripheral surface (inner surface of the side wall) of the first swirl flow forming portion **50**A is the curved first curved surface **501**A, and an inner peripheral surface (inner surface of the side wall) of the second swirl flow forming portion **50**B is the curved second curved surface **501**B. Thereby, the first swirl flow forming portion **50**A can form the first swirl flow **5**A more suitable for stirring, and the second swirl flow forming portion **50**B can form the second swirl flow **5**B more suitable for stirring. Accordingly, the mixture M**7** can be stirred and loosened further satisfactorily in the supply section **5**.

The present disclosure is not limited to the above configuration, and the inner peripheral surfaces of the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B may have a plurality of flat surfaces, or may have a configuration in which a curved surface and a flat surface are combined.

In addition, as described above, the first curved surface **501**A and the second curved surface **501**B have a shape that is symmetrical with respect to the boundary portion **56** between the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B. Thereby, the shapes of the first swirl flow **5**A and the second swirl flow **5**B can be formed in a well-balanced manner, and the strength and the swirl speed of both swirl flows can be made more uniform. Accordingly, the mixture M**7** can be evenly and efficiently stirred and loosened in the supply section **5**.

The present disclosure is not limited to the above configuration, and the first curved surface **501**A and the second curved surface **501**B may have a shape that is asymmetrical with respect to the boundary portion **56**.

In addition, as described above, the end part **58**, which is a downstream part, of the supply pipe **57** is coupled to the boundary portion **56** between the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B. Thereby, the mixture M**7** supplied from the supply pipe **57** is divided into the first swirl flow forming portion **50**A and the second swirl flow forming portion **50**B equally or as close to equal as possible, so that the first swirl flow **5**A and the second swirl flow **5**B can be formed in a well-balanced manner. Accordingly, the mixture M**7** can be evenly stirred and loosened in the supply section **5**.

The present disclosure is not limited to the above configuration, and the supply pipe **57** may have a configuration in which the supply pipe **57** is branched into two parts in the middle thereof, a downstream end of one of the branched pipes is coupled to the first swirl flow forming portion **50**A, and a downstream end of the other branched pipe is coupled to the second swirl flow forming portion **50**B. In this case, a coupling direction and a coupling portion of each branched pipe to the chamber **50** are not particularly limited, and for example, a configuration may be adopted in which each branched pipe is coupled to the chamber **50** from the −x axis side or the +x axis side, or a configuration may be adopted in which one branched pipe is coupled along the first curved surface **501**A and the other branched pipe is coupled along the second curved surface **501**B.

As described above, although the dispersion device and the accumulation device of the present disclosure are described based on the illustrated embodiment, the present disclosure is not limited to this, and the configuration of each section can be replaced with any configuration having the same function. In addition, in the present disclosure, other any components may be added to the above-described embodiment.

What is claimed is:

1. A dispersion device comprising:

a supply section having a supply pipe for supplying a material containing fibers together with air, and a chamber that has a first swirl flow forming portion forming a first swirl flow of the air containing the material and a second swirl flow forming portion communicating with the first swirl flow forming portion and forming a second swirl flow in a direction opposite to the first swirl flow of the air containing the material and that is coupled to the supply pipe;

a dispersion section that has a casing formed with a discharge port for discharging the material, stirs the material in the casing, and discharges and disperses the material from the discharge port into air; and

a coupling section having a communication port through which the first swirl flow forming portion and the second swirl flow forming portion communicate with the casing.

2. The dispersion device according to claim **1**, wherein

the dispersion section has a stirring member installed in the casing and rotating around a rotation axis.

3. The dispersion device according to claim **2**, wherein

the communication port has an elongated shape extending in a first direction parallel to the rotation axis.

4. The dispersion device according to claim **2**, wherein

the first swirl flow forming portion and the second swirl flow forming portion are disposed side by side in a first direction parallel to the rotation axis.

5. The dispersion device according to claim **1**, wherein

an inner peripheral surface of the first swirl flow forming portion is a curved first curved surface, and an inner peripheral surface of the second swirl flow forming portion is a curved second curved surface.

6. The dispersion device according to claim **5**, wherein

the first curved surface and the second curved surface have a shape symmetrical with respect to a boundary portion between the first swirl flow forming portion and the second swirl flow forming portion.

7. The dispersion device according to claim **1**, wherein

an end part, which is a downstream part, of the supply pipe is coupled to a boundary portion between the first swirl flow forming portion and the second swirl flow forming portion.

8. An accumulation device comprising:

a supply section having a supply pipe for supplying a material containing fibers together with air, and a chamber that has a first swirl flow forming portion forming a first swirl flow of the air containing the material and a second swirl flow forming portion communicating with the first swirl flow forming portion and forming a second swirl flow in a direction opposite to the first swirl flow of the air containing the material and that is coupled to the supply pipe;

a dispersion section that has a casing formed with a discharge port for discharging the material, stirs the material in the casing, and discharges and disperses the material from the discharge port into air;

a coupling section having a communication port through which the first swirl flow forming portion and the second swirl flow forming portion communicate with the casing; and

an accumulation section accumulating the material dispersed by the dispersion section.

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